

Software Engineering Lab 1

Name: Likhith S
SRN: PES1UG24CS720
Section: L

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall maintain a time-credit balance for each member, crediting 1 hour upon verified completion of a teaching session.	High	Pass: Time credit transfers from learner to mentor after session sign-off. Fail: Member with 0 credits books a session.	Core mechanism that makes the time-banking exchange work.
FR-002	Functional	The system shall allow a member to list the skills they can teach, including a title, description, and available time slots.	High	Pass: A listed skill appears in search results within 5 seconds of publishing. Fail: An unpublished or deleted listing still appears in search.	Enables discovery - the marketplace has no supply side without this.
FR-003	Functional	The system shall allow a learner member to request a session with a mentor for a listed skill, specifying a preferred time slot.	High	Pass: Mentor receives a session request notification with learner name, skill, and time slot within 1 minute of submission. Fail: Request is submitted with no time slot selected.	Booking is the core transaction of the platform.
FR-004	Functional	The system shall require both learner and mentor to submit a mutual rating (1-5 stars) and optional comment within 48 hours of session completion before time credit is finalized.	Medium	Pass: Time credit is marked "finalized" only after both parties submit a rating. Fail: Credit finalizes with only one party having rated.	Ties the feedback validation from the scenario directly to the credit system, preventing unverified sessions from paying out.
FR-005	Functional	The system shall allow either party to flag a session as a no-show or dispute within 24 hours of the scheduled session time, freezing the associated credit transfer pending admin review.	Medium	Pass: Flagging a session sets its status to "disputed" and blocks credit transfer. Fail: A flagged session's credits transfer automatically without review.	Protects members from bad actors gaming the credit system; makes the platform trustworthy.
NFR-001	Nonfunc. (Performance & Security)	The session scheduling calendar must sync seamlessly with Google Calendar and Outlook via iCal export feeds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given requirement - ensures scheduling fits into members' existing workflows.
NFR-002	Nonfunc. (Privacy)	The system shall never expose a member's real contact details (phone/email) to another member; all in-platform communication shall route through an anonymized messaging layer.	High	Pass: A member's raw phone number/email is absent from all API responses and UI views visible to other members. Fail: Contact details appear anywhere outside the platform's own messaging feature.	Peer-to-peer exchange platforms carry real safety/privacy risk when strangers can see each other's contact info directly.